

SIMATS ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences

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COURSE NAME: COMPUTER NETWORK

1. Multiplexing vs. Demultiplexing :

Multiplexing	Demultiplexing
1. Performed at the **sender's transport layer**.	1. Performed at the **receiver's transport layer**.
2. Collects data from **multiple application sockets**.	2. Receives segments/datagrams from the network.
3. Adds **source and destination port numbers**.	3. Uses **port numbers** to identify the destination socket.
4. Encapsulates application data into **TCP segments or UDP datagrams**.	4. Extracts the application data from received segments/datagrams.
5. Allows multiple applications to share the same network connection.	5. Delivers each data unit to the **correct application process**.
6. Example: browser, email, and video applications can send data simultaneously.	6. Example: port 443 data is delivered to the appropriate HTTPS application.

2. TCP Sockets vs. UDP Datagrams :

TCP Sockets	UDP Datagrams
1. TCP is **connection-oriented** and establishes a connection before communication.	1. UDP is **connectionless** and sends data without establishing a connection.
2. Provides a continuous **byte-stream service**.	2. Provides a **message/datagram-oriented service**.
3. Maintains connection state using sequence numbers, acknowledgments, and other control information.	3. Maintains minimal transport-level state, making it lightweight.
4. Provides **reliable and ordered delivery**.	4. Does not guarantee delivery or ordering.
5. Lost packets are detected and retransmitted by TCP.	5. Lost datagrams are not automatically retransmitted by UDP.

6. Suitable for **HTTPS, file transfer, and reliable applications**.	6. Suitable for **DNS, gaming, streaming, and real-time applications** where low overhead is important.

3. Multicast vs. Unicast/Broadcast :

Multicast	Unicast / Broadcast
1. Multicast sends data to a **specific group of receivers**.	1. Unicast sends data to one receiver, while broadcast sends it to all applicable hosts.
2. One stream of UDP packets can serve multiple subscribed receivers.	2. Unicast may require separate copies for multiple receivers.
3. Receivers must **join a multicast group** to receive the traffic.	3. Broadcast generally does not require individual hosts to join a group.
4. Reduces unnecessary duplication of identical data.	4. Multiple unicast copies can consume more bandwidth.
5. Efficient for **one-to-many applications** such as IPTV and live distribution.	5. Unicast is common for normal client-server communication; broadcast can create unnecessary traffic.
6. Network devices can replicate multicast traffic only where required.	6. Broadcast can send traffic to many hosts, including hosts that do not need the data.

4. TCP Congestion Control vs. Traffic Management :

TCP Congestion Control	Traffic Management
1. TCP congestion control is mainly performed by **end hosts**.	1. Traffic management can be performed by **routers and network devices**.
2. It detects congestion using packet loss, delay, acknowledgments, and related signals.	2. Routers can monitor and manage traffic in their queues.
3. **Slow Start** initially increases the sending rate gradually.	3. **QoS** can prioritize important traffic classes.

4. **AIMD** increases the rate and reduces it when congestion is detected.	4. **RED** can detect growing queues and drop/mark packets early.
5. TCP reduces its sending rate when network congestion is inferred.	5. **Traffic policing/shaping** can control the amount and rate of traffic entering a network.
6. It is primarily an **end-to-end adaptive mechanism**.	6. It is a **network-assisted mechanism** that manages traffic and resources inside the network.

5. Single-Server vs. Multi-Server Queuing Models

M/M/1 – Single Server	**M/M/c – Multi-Server**
1. Has **one server** serving customers from a queue.	1. Has **c parallel servers** serving customers from a common queue.
2. Arrivals follow a **Poisson process** with rate (λ).	2. Arrivals also follow a Poisson process with rate (λ).
3. Service times follow an **exponential distribution** with rate (μ).	3. Each server generally has an exponential service rate (μ).
4. Traffic intensity is ($\rho = \lambda/\mu$), and stability requires ($\rho<1$).	4. Traffic intensity is ($\rho = \lambda/(c\mu)$), and stability requires ($\rho<1$).
5. Average queue waiting time is ($W_q = \lambda/[\mu(\mu-\lambda)]$).	5. Average waiting time is based on the **Erlang-C** model: ($W_q=P_W/(c\mu-\lambda)$).
6. Waiting time can become large when the single server is highly utilized.	6. Multiple servers provide pooled capacity and generally reduce waiting time for the same arrival load.

6. Priority Queues vs. Network of Queues

Priority Queuing	Network of Queues
1. Packets are classified into **different priority levels**.	1. Consists of **multiple interconnected queues**.
2. High-priority packets are served before lower-priority packets.	2. The output of one queue can become the input to another queue.
3. Voice or real-time traffic can be given higher priority than background traffic.	3. Packets can pass through several processing stages or network nodes.
4. It focuses mainly on **service order**.	4. It focuses mainly on **traffic flow between queues**.
5. Low-priority packets may experience longer delays or starvation.	5. Delays can accumulate as packets move through multiple queues.
6. Useful for **QoS-based packet scheduling**.	6. Networks can be **open** (traffic exits) or **closed** (traffic circulates within the system).

7. QUIC Protocol vs. TLS 1.3

QUIC	TLS 1.3
1. QUIC is a modern **transport protocol** that operates over UDP.	1. TLS 1.3 is a **security and cryptographic protocol**.
2. Provides **multiplexed independent streams**.	2. Provides authentication, encryption, and secure key establishment.
3. Uses independent streams to avoid transport-level **Head-of-Line blocking between streams**.	3. TLS itself does not provide transport-layer stream multiplexing.
4. Provides reliability, congestion control, acknowledgments, and connection migration.	4. Provides **confidentiality, integrity, and authentication**.
5. Uses **Connection IDs** so connections can survive changes in network address.	5. TLS 1.3 provides the cryptographic handshake used by QUIC.

6. QUIC + TLS 1.3 + UDP forms the foundation for **HTTP/3**.

6. With traditional HTTPS/HTTP/2, TLS 1.3 operates above **TCP**, while QUIC integrates TLS 1.3 into its connection establishment.